

TITLE

IntelliCane : A Smart Embedded Cane with Safety features for the Elderly and Visually Impaired

AUTHORS

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ABSTRACT

This paper presents the design and implementation of IntelliCane, a smart embedded assistive cane developed to enhance safety, mobility, and independence for elderly users and the visually impaired. The system features a voice recognition module that enables users to select operating modes tailored to either elderly or visually impaired users, along with obstacle detection and distance measurement for safe navigation, supported by vibration-based feedback. A GPS module provides real-time location tracking, while an MPU6050 motion sensor detects abnormal movements related to emergencies or potential cane theft, triggering alerts to family members or caregivers. Additional safety features include an automatic lighting system for low-light environments, an onboard emergency alert button, and a portable rechargeable battery pack for continuous operation. The successful integration of sensing, feedback, communication, and power subsystems demonstrates the practicality of IntelliCane as a compact mechatronic assistive device, with experimental results indicating reliable obstacle detection and timely emergency response.

1. INTRODUCTION

Elderly people and individuals with visual impairments often face difficulties when moving around safely and independently. Traditional mobility aids such as the white cane can only detect obstacles when they are touched, which means hazards at head level, uneven paths, or moving objects may not be detected in time. This can lead to collisions, falls, and injuries, especially in unfamiliar places or crowded areas.

Another major problem is the lack of emergency support in conventional walking aids. If a user falls, feels unwell, or becomes disoriented, there is usually no quick way to send their location or request help. This delay can be dangerous, particularly when the user is alone or in poor lighting conditions. Some existing smart canes try to solve this issue, but many are expensive, slow to respond, or too complex to be reliable for daily use.

In addition, many assistive devices focus on advanced features without ensuring fast and reliable responses for safety. When obstacle detection, alerts, and decision-making are handled by a single processing unit, delays may occur, increasing the risk of accidents. Poor power management can also reduce operating time, making the device less suitable for long-term daily use. To address these limitations, this project aims to develop a Smart Cane, also referred to as IntelliCane, as a multi-functional Cyber-Physical System that integrates modern sensing, communication, and control technologies. The primary objective is to enhance user safety and independence by enabling real-time obstacle detection and distance measurement, implementing an emergency alert system that utilizes GPS and IMU (MPU6050) sensors for fall or distress detection, providing intuitive haptic feedback through vibration modules to warn users of nearby hazards, and improving visibility and personal security through an automatic lighting system that activates under low-light conditions.

Modern mechatronics system integration is commonly based on a multi-tier architectural approach that separates time-critical control tasks from high-level computational processing. In this project, the system architecture is divided into two main processing layers to optimize performance and reliability. The ESP32 microcontroller functions as the deterministic control unit, responsible for handling time-sensitive operations such as acquiring data from sensors including LIDAR and the MPU6050, as well as generating immediate input–output responses such as vibration and buzzer alerts. In parallel, a Raspberry Pi 5 is incorporated as the main processing unit to manage computationally intensive tasks such as voice recognition and potential artificial intelligence–based object classification. Distance sensing is achieved using Time-of-Flight principles through LIDAR technology, which provides higher accuracy and consistency compared to conventional ultrasonic sensors. For emergency communication, the

integration of a GPS module enables real-time location tracking, allowing the system to transmit the user's location to caregivers via wireless communication protocols during emergency situations.

It is hypothesized that the implementation of a dual-processor architecture using the ESP32 and Raspberry Pi 5 will minimize response latency for safety-critical functions while simultaneously supporting advanced user interaction features. The expected outcome is a reliable hazard detection system capable of identifying obstacles within a range that allows haptic feedback to be delivered within 50 ms, thereby giving the user sufficient reaction time. In addition, the emergency alert mechanism is expected to transmit accurate GPS coordinates with a positional error of approximately 3 to 4 meters when activated through manual input or automatic fall detection. Finally, the system is designed to achieve stable and sustainable operation through the use of a rechargeable portable battery pack, ensuring consistent power delivery to all integrated modules. By integrating a dual-processor mechatronics architecture with real-time sensing, intelligent task partitioning, and robust emergency communication, the proposed IntelliCane directly addresses the limitations of traditional mobility aids by enabling safer, faster, and more reliable navigation for elderly and visually impaired users.

2. METHODOLOGY

2.1 Project Development

Navigating unfamiliar environments independently is a significant challenge for visually impaired individuals. To address this, the smart cane utilizes 5 systems embedded in the product. The first figure (a) shows the user interacting with the graphic user interface to pick an option between elderly or visually impaired. Figure (b) shows the second system integration that has advanced object detection sensors to identify obstacles in the user's path, providing immediate vibratory feedback to prevent accidents. Furthermore, the third system that integrates GPS tracking and navigation to lower the risk of getting lost. Lastly, the combination of system 4 and 5, that has an emergency alert feature that allows the user to instantly send their location to caregivers, ensures safety and peace of mind by the press of a button or when the cane is stolen, and light that will activate automatically when the surrounding is dark. Figure (c), indicates the flow chart of all the system that represent the work principle of IntelliCane.

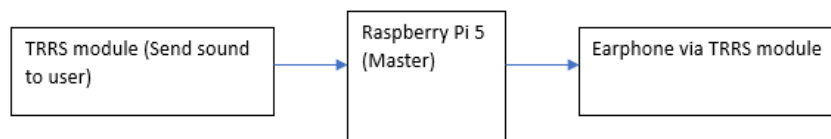


Fig. A Block diagram of system 1 for GUI

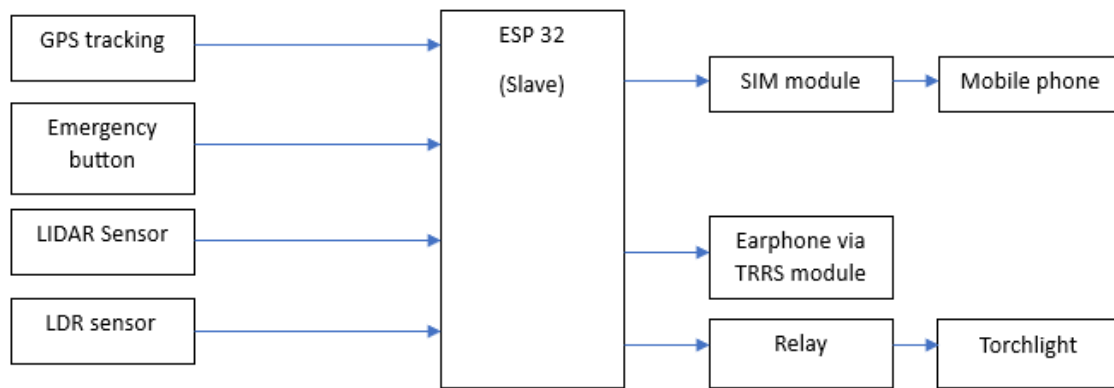


Fig. B Block diagram of system 2 (object detection), system 3 (GPS navigation), system 4 (emergency button and gyro sensor) and system 5 (auto light system)

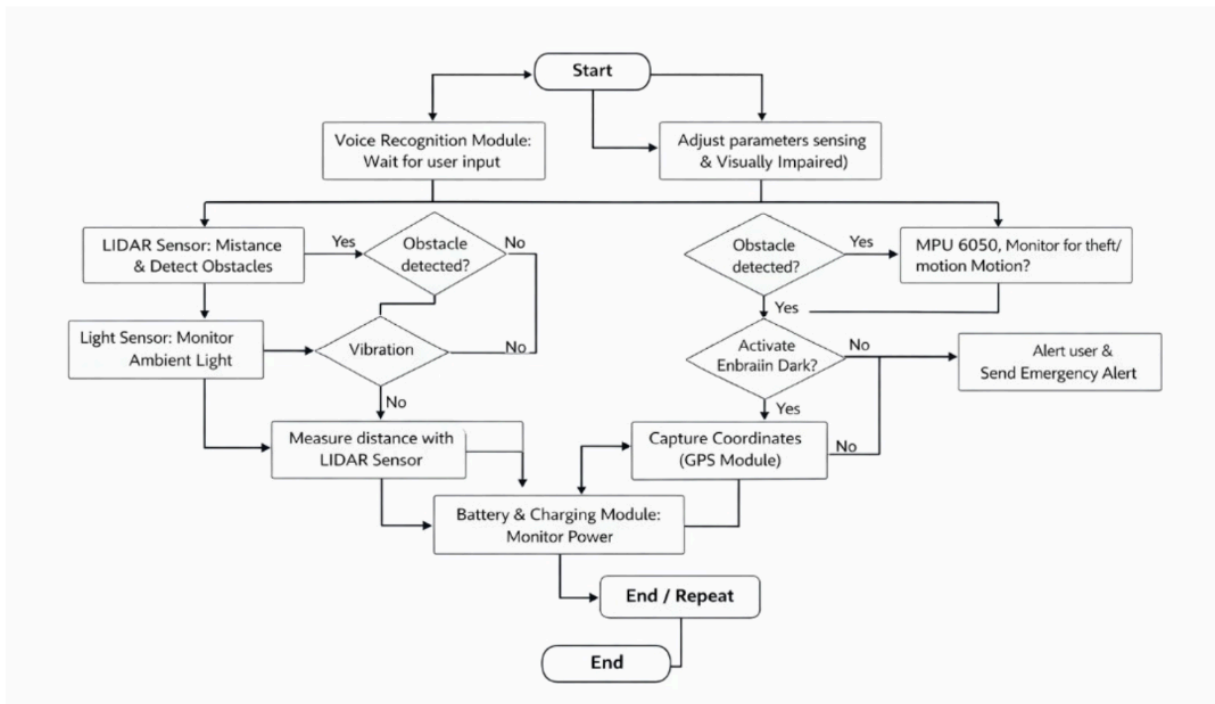


Fig. C Flowchart explaining the work principle of IntelliCane

